[INCH-POUND]
A-A-59273
June 18, 1998

COMMERCIAL ITEM DESCRIPTION

GENERATOR, SIGNAL
300 KHZ TO 2 GHZ

The General Services Administration (GSA) has authorized the use of this Commercial Item Description for all federal agencies.

1. SCOPE.

This Commercial Item Description (CID) outlines the minimum requirements for a programmable signal generator operating in the frequency range of 300 Kilohertz (KHz) to 2 Gigahertz (GHz). The signal generator offered to meet the requirements of this CID shall be commercially available test equipment capable of being modified to meet the following description. Only those manufacturers that meet or surpass the requirements outlined in this CID are acceptable.

2. SALIENT CHARACTERISTICS.

2.1. Safety and environmental. The signal generator shall be solid state and enclosed to provide protection from electrical shock. The signal generator shall meet all safety and environmental requirements as specified in MIL-PRF-28800F for Class 3 test equipment and must be capable of being operated as bench-top or rackmount equipment.

2.2. Electrical power sources. The signal generator shall operate from nominal commercial, military, and shipboard power sources of 110/220 volts AC (±10 percent) at line frequencies of 50 to 440 Hz. Power consumption shall not exceed 500 Volt Amps (VA). Transient state conditions shall meet MIL-PRF-28800F requirements for Class 3 test equipment.

2.3. Temperature requirements. The signal generator shall meet its performance and accuracy requirements as Class 3 test equipment in an operating environment of plus 10 degrees to

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plus 50 degrees Celsius as stated in MIL-PRF-28800F. After storage in a non-operating environment between minus 50 and plus 60 degrees Celsius, the signal generator shall function properly and meet its performance and accuracy requirements.

2.4. Preconditioning requirement. Each deliverable signal generator shall be capable of being subjected to the environmental screening of burn-in and thermal cycling, with thermal cycling being accomplished in the sample testing phase.

2.5. Calibration and maintenance adjustments. The calibration procedures for this signal generator shall meet MIL-PRF-28800F, paragraph 3.12., concerning calibration procedures. Calibration adjustments shall be provided by the variable value components, which are adjustable by the use of simple means. The calibration by substitution of selected components or parts is unacceptable unless specifically approved in advance. The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules. The calibration interval shall be a period of one year or greater based on an operating time of 2,000 hours.

2.6. Reliability. The Mean Time Between Failure (MTBF) of this signal generator shall be 2,000 hours. The lower test MTBF shall be 2,000 hours and the upper test shall be 4,000 hours.

2.7. Electromagnetic Interference. This test equipment shall comply with the electromagnetic interference (EMI) requirements as specified in MIL-STD-461.

2.8. Performance characteristics.

2.8.1. Frequency range. The signal generator shall function in the frequency range of 300 KHz to 2 GHz with a resolution of 1 Hz.

2.8.2. Frequency stability. The generator shall have a frequency stability of 5×10^{-10} /day.

2.8.3. Spectural purity:

2.8.3.1. Single Side Band (SSB) phase noise. The signal generator shall have a SSB phase noise of -130 dBC/Hz at 1 GHz and 20 KHz offset.

2.8.3.2. Non-harmonics spurious. The signal generator shall have a non-harmonics spurious of -100 dBc for center frequencies of 300 MH to 1 GHz, and -94 dBc for center frequencies of 1 GHz to 2 GHz with 10 KHz offset.

2.8.3.3. Harmonic. The signal generator shall have a harmonic of -27 dBc when the output is less than +8 dBm.

2.8.3.4. Subharmonic. The signal generator shall have a subharmonic of –60 dBc for frequency range of 500 MHz to 1.2 GHz, and -40 dBc for a frequency range of 1.2 to 2 GHz.

2.8.4. Output level. The signal generator shall have an output of +13 to -127 dBm, a resolution of 0.1 dB, absolute accuracy of ±1 dB, and a reverse power protection of 25 Watts.

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2.8.5. Amplitude Modulation (AM). The AM mode of this signal generator shall have a depth of 0 to 99 percent, a resolution of 0.1 percent, a bandwidth of dc to 50 KHz, and a distortion of 5 percent.

2.8.6. Frequency Modulation (FM). The FM mode of this signal generator shall have a maximum peak deviation of 1 MHz for the frequency range 500 MHz to 1 GHz, 2 MHz for the frequency range 1 GHz to 2 GHz, a resolution of 2.5 percent of setting, a bandwidth of dc to 100 MHz, and a distortion of 5 percent.

2.8.7. Pulse modulator. The pulse modulator mode of this signal generator shall have an ON/OFF ratio of 50 dB for frequency range of 500 MHz to 1.2 GHz, and 80 dB for frequency range of 1.2 GHz to 2.0 GHz; and a rise and fall time of 100 nanoseconds.

2.8.8. Internal modulation source. The signal generator shall have an internal modulation source for the following waveforms: sine, square, triangular, and sawtooth.

2.9. Signal generator features required as options.

2.9.1. Rackmount Option. If specified in the contract or order, the signal generator shall be delivered with an attachable rackmount kit which includes the necessary accessories and hardware to interconnect all front and rear panel input and output ports and include any filler panels to provide the preferred height and width, if required. Conversion to a rackmount configuration shall be accompanied by the use of a kit that converts the enclosure but will not prevent return to the original configuration.

2.9.2. Interface Option. The signal generator shall be provided with an IEEE-STD-488 interface and connector which meet requirements stated in MIL-PRF-28800F, paragraph 3.4.1.

2.9.3. High Stability Timebase Option. The high stability timebase option shall provide the following minimum capabilities:

2.9.3.1. Internal reference oscillator at frequency of 10 MHz, temperature 0 to 50 degrees C. with less than $\pm 2 \times 10^{-10}$ /degrees C.

2.9.3.2. Aging rates less than $\pm 5 \times 10^{-10}$ /day.

2.9.3.3. Output level minus less than 0.2 VRMS into 50 ohms.

3. REGULATORY PROVISIONS.

In accordance with Section 23.403 of the Federal Acquisition Regulation, the Federal Government’s policy is to acquire commercial items composed of the highest percentage of recoverable materials practicable consistent with maintaining a satisfactory level of competition and without adversely affecting performance requirements or exposing the supplier’s employees to undue hazards from the recovered materials.

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4. QUALITY ASSURANCE PROVISIONS.

4.1. Product conformance. The signal generator provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The Federal Government reserves the right to require proof of such conformance.

4.2. Submitted products. The signal generator, submitted for bid sample testing, shall be representative of the commercial off-the-shelf product. All features, performance, and environmental protection of the product tested during bid sample testing shall be provided under any resulting contract or purchase order.

4.3. Year 2000 compliance. This signal generator shall accurately process date ranges and time data from, into, and between the 20th and 21st centuries and specifically, the years 1999, 2000, 2001 and leap year calculations. This compliance includes comparison and sequential date/time data. The signal generator shall also accurately process date and/or time data if such data is received or exchanged from other equipment or sources.

5. PACKAGING.

5.1. Preservation, packaging, labeling, and marking. The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order. Each signal generator shall have an identification plate permanently affixed in a readable location which states the manufacturer's name, the model number, the equipment name, and its serial number.

6. NOTES.

6.1. Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified in conversion tables contained in the latest edition of Federal Standard No. 376 and all other requirements of this CID are met. If the manufacturer elects to use metric dimensions as the basis upon which to derive or base a mechanical feature, a request shall be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

6.2. Source of document publications. Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Defense Automated Printing Service, 700 Robbins Ave., Bldg. 4D, Philadelphia, PA 19111-5094. Phone 215/697-2179.

6.3. Possible acceptable products. Following is a list of assigned National Stock Numbers (NSNs), although not inclusive of all possible products, which may meet the requirements outlined in this CID.

Manufacturer	Model No.	National Stock Number
Hewlett-Packard Co.	8643A-001-002	6625-01-434-7691
Giga-tronics, Inc.	6082A-130	6625-01-443-4085

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